COMP 3625 Assignment 2

Supervised learning & decision making

This assignment will become visible in D2L *after* you join a D2L group.
You **must** work in a group of 3~4 for this assignment. Individual assignments by permission only.

Demo Sign-up Sheet Here

The Assignment

Investors have given you $100 million to start an insurance company. You are tasked with writing a program that will select customers to take on as clients.

You have acquired a historical dataset of insurance customers. It includes 21 features of each customer including their coverage amount, and indicates whether they made a claim. You can analyse this data to build your program.

At evaluation time, your program will be given a dataset of new, potential customers (but you won't know which of them will claim). Your program must:

1. Estimate the probability that each potential customer will make a claim

2. Set a premium price for each customer, that both offers them a utility advantage (over going uninsured) and makes your company an expected profit.

3. Choose which customers to take on, such that the total coverage amount does not exceed your $100 million startup fund.

Assumptions

Make the following assumptions to simplify the problem:

- Assume that when a claim is made, it is always for the full coverage amount.

- Assume that customers will make at most 1 claim. So the problem is simply to estimate which customers will claim, and which won't.

- You must have enough cash on hand to cover all your insurance policies. That is, the total coverage amount for all your customers can't be more than $100 million. This limits the number of customers you can take on.

- Assume premiums are a one-time charge (not a monthly fee as in real life)

- Assume all customers derive utility from money according to the following utility curve [1]:

  ```
  U($) = -263.31 + 22.09 ln($ + 150,000)
  ```

  where $ is the monetary amount in dollars.

Requirements

1. Your program must involve:

   - A supervised learning component, to predict the probability that a given customer will make a claim.

   - At least one search.
     A few example use cases for search:
     i. to search for optimal model parameters.
     ii. for feature selection. In most practical problems, some features are redundant or irrelevant. Selecting the best features may improve generalization.
     iii. to find the best set of customers to take on.

   You must **write at least one of these components (the supervised learning or the search) "from scratch"**[2].

[1] this curve was suggested in "decision under uncertainty" by Grayson,1960
[2]Well, not *completely* from scratch - you can use numpy/pandas objects or other low-level building blocks.

2. Your premiums must offer a utility benefit to the customer. That is, given the above utility curve and your estimate of the customer's probability of making a claim, your premium must be set to provide better utility than the lottery of going uninsured.
   I need to see this calculation being performed explicitly in your code.

    - Beyond this requirement, you're free to set the premiums however you like - but see the "friendly competition" described below.

3. I will evaluate your program by calling the `set_premium` function. This function can load saved models and do any other computations necessary, but it must run in ~60s or less.
   See the function docstring for more info on how to implement it.

Starter code

The data folder contains two files:

- customer_data.csv - data on each of 100,000 customers
- customer_claims.csv - data indicating which customers made claims

Deliverables

Code and written responses

After you finish coding, tackle the questions in written_responses.md

Demos and friendly competition

On an assigned day, you will briefly explain your program to your instructor, explaining and justifying your design decisions.

I will run everyone's programs on a new dataset of potential customers, and we'll note the premiums being offered to each customer by each team. Each customer will accept the **lowest** offer. We'll measure how much money each team made from the customers they attracted.

Hints and thoughts

- Note that sklearn models (and other libraries that follow an sklearn-like API) have a “predict_proba” method that returns the estimated probabilities of each instance belonging to each class.

- Think about how you’ll decide which customers to take on, up to the $100M limit. What objective will you try to optimize? Maximum expected profit? Minimum risk? Maximum number of clients? Something else?

- Remember that you are now dealing with the *negative* side of the logarithmic utility curve: the customer is weighing the *loss* of paying you a premium, vs the *expected loss* of the lottery of going uninsured.

- Don’t overthink the setting of premiums. A simple rule, like “20% over the expected cost of the policy” is totally reasonable. (Just remember you *must* offer a utility benefit to the customer).

Evaluation

Per the MRU grading table in the course outline, an “A” solution represents “Excellent performance, showing comprehensive understanding of subject matter.”
An “A” submission for this assignment will:

- Perform well & meet requirements
- Be well structured, easy to read and understand
- Show significant understanding of supervised learning and the principle of maximum expected utility.
- Have clear & concise written responses that clearly demonstrate understanding
- Include a clear demonstration that explains what the program is doing and why you built it that way
- Include git commit history that shows a progression of work contributed by all group members.

A “C” submission represents “Basic understanding of subject matter”, and might:

- Include sloppy, difficult-to-read code. Possibly one big dump of code into github on the due date.

- Not perform much better than a trivial or brute-force solution
- Include written responses that are vague, surface-level, off-topic, or rambling.
- Include a demo in which choices are not justified or a lack of effort is apparent.
    - If it becomes clear at any point that you submitted code you do not understand, you will most likely receive a **zero grade**!

Submission

Enter a **github repository url** into the D2L assignment.
Make sure you add me (echalmers) as a collaborator on the repo, so I have read access.